When Does a Foundation Model Help? Process-Specific Reliability of Machine-Learned Interatomic Potentials

Yi Cao[1], Peter Mastracco[1], Paulette Clancy[1*]

[1]Department of Chemical and Biomolecular Engineering, Johns Hopkins University, 3400 N. Charles Street, Baltimore, 21218, Maryland, USA.

*Corresponding author(s). E-mail(s): pclancy3@jhu.edu;
Contributing authors: ycao73@jh.edu; pmastra2@jh.edu;

Abstract

Foundation models promise to make accurate atomistic simulation accessible with little material-specific training. Their practical value, however, depends on whether they reproduce the process for which a simulation is needed. Here we use Cr-doped Sb_2Te_3 to compare specialist, zero-shot and fine-tuned interatomic potentials across thermal configurations, dopant migration and collective layer sliding. Fixed-geometry energies, model-relaxed pathways and molecular dynamics expose complementary failure modes. Target adaptation improves force accuracy and selected migration-energy comparisons, yet does not ensure stable dynamics or reliable energies under stronger distortions. Conversely, zero-shot checkpoints sustain longer trajectories while remaining poorly calibrated on some migration configurations. Relaxing a pathway can lower its apparent barrier without recovering the reference energy profile. Structural projections and error surrogates help characterize these differences, but do not establish their microscopic cause. These results motivate process-specific model selection: a useful potential must be tested both on the configurations relevant to an observable and on the configurations its own forces generate. Foundation initialization is therefore a resource to validate and adapt, rather than a guarantee of downstream reliability.

Keywords: machine-learned interatomic potentials (MLIPs), foundation models, fine-tuning, migration, distribution shift, model validation

1 Introduction

Foundation models are changing how atomistic simulations are built. Rather than assembling a new potential for every material, researchers can start from a model trained across diverse chemical environments and use it directly or adapt it with a smaller reference dataset. Machine-learned interatomic potentials (MLIPs) have made this prospect especially attractive: they connect first-principles energies and forces to simulations at scales that are difficult to reach with density functional theory (DFT) alone [1–14]. Broadly trained potentials such as MACE-MP and CHGNet extend this approach across materials [15–21]. The resulting choice is increasingly practical: should a researcher use a foundation checkpoint, fine-tune it, or train a specialist?

For a materials scientist, the answer must ultimately be expressed in terms of a physical process. A model may reproduce typical thermal configurations yet misrepresent the short-lived coordination changes that control dopant motion. It may remain numerically stable while exploring an inaccurate energy landscape. Conversely, a small energy error on a supplied transition path does not guarantee that the model's forces will generate that path. These distinctions matter because activated motion depends on relative energies along rare configurations, while a simulated trajectory repeatedly uses the model's own predictions to determine what it sees next. Average error over a test set and reliability in a target process are therefore different questions [22–26]. Broad materials benchmarks such as Matbench and JARVIS-Leaderboard assess property-prediction performance at scale, while generative benchmarks examine a different task [27–29].

Recent work establishes both the promise and the conditional nature of transfer. Alghamdi and colleagues show that zero-shot and lightly fine-tuned MACE models can reproduce Li diffusion observables in LiF; Bheemaguli and colleagues distinguish migration-energy accuracy from transition-path geometry across battery materials; and Deng and colleagues identify systematic softening of universal potentials on several out-of-distribution tasks [30–32].[R1-A7] These studies make process-aware testing essential. They also suggest that the success of adaptation depends on which parts of the energy landscape the target data constrain [33–39]. What remains useful to resolve is how the verdict changes when the same material is examined through different processes and different evaluation procedures.

We address this question using Cr-doped Sb_2Te_3. Its quintuple layers comprise a Te–Sb–Te–Sb–Te stack, separated by van der Waals gaps. A Cr atom moving within a gap, approaching a layer or entering its interior encounters different coordination environments. Translating one layer relative to another introduces a collective registry change. This combination supplies a compact set of physically distinct probes within one host (Fig. 1a,c), without treating a change of material as the only form of distribution shift. More broadly, layered 2D materials are studied across spintronic, quantum and energy applications, where stacking, intercalation and magnetic doping provide routes to tune their properties [40–55]. Cr-doped Sb_2Te_3 is also an established platform for studying the interplay of transition-metal doping and layered topological materials [56–67].

The comparison separates two sources of difficulty. The first is a change in the configurations being evaluated, from thermal fluctuations to migration and sliding.

The second is a change in how configurations are obtained: a common reference path tests energy assignments, whereas an MLIP-relaxed band or an MD trajectory tests the feedback between predictions and subsequent configurations. This distinction makes it possible to ask what a foundation model helps with, where adaptation helps, and which apparent improvements fail to carry over to another use.

Here we test whether foundation initialization, target-data adaptation and specialist training remain reliable across thermal configurations, dopant migration and collective sliding in one host. We connect held-out accuracy to both common-geometry scoring and model-generated configurations, keeping these evaluation questions distinct. The goal is to identify process-specific strengths and failure modes, not to assign a universal ranking to training routes.

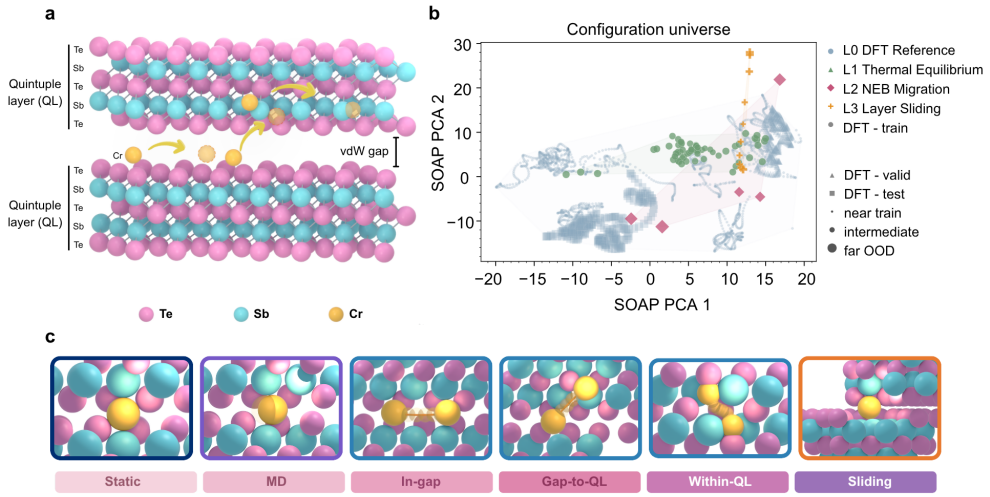


Fig. 1 A material contains several distinct tests of an interatomic potential. a, Cr-doped Sb_2Te_3, showing the Te–Sb–Te–Sb–Te quintuple layers (QLs), the van der Waals gap and representative dopant displacements. Te, Sb and Cr are pink, cyan and yellow, respectively. **b,** SOAP-based principal-component visualization of reference configurations, thermal trajectories, migration images and layer-sliding configurations. Symbols distinguish configuration sources and reference split assignments; the projection is a descriptive view of sampled configurations, not a calibrated measure of extrapolation risk. **c,** Local views of static configurations, molecular dynamics, in-gap migration, gap-to-QL motion, within-QL motion and sliding, in that order. The task categories organize the tests; their ordering does not define a universal or quantitatively monotonic distribution-shift scale.

2 Results

2.1 Pointwise accuracy gives different answers for energies and forces

The held-out comparison shows that energy and force accuracy do not produce the same model ordering (Fig. 2a–h). Fine-tuning lowers test force error relative to scratch training in both target-data regimes (Fig. 2e–h), while the energy comparison does not follow the same ordering (Fig. 2a–d). On the 600 K test partition, FT-600K has an energy RMSE of 40.68±4.88 meV·atom^{-1} and a force RMSE of 259.68±5.83 meV·Å^{-1}, compared with 30.48 ± 3.48 meV·atom^{-1} and 343.55 ± 36.27 meV·Å^{-1} for Scratch-600K (Supplementary Table S1). These are means and sample standard deviations over training seeds. Adaptation improves the force observable in these comparisons, while energy agreement favours another route.

The plotted energy correlations are within-cluster diagnostics, whereas the reported uncentred RMSE retains absolute errors. Neither pointwise measure tests how accurately a model describes a rare coordination change or what configurations its own forces generate.

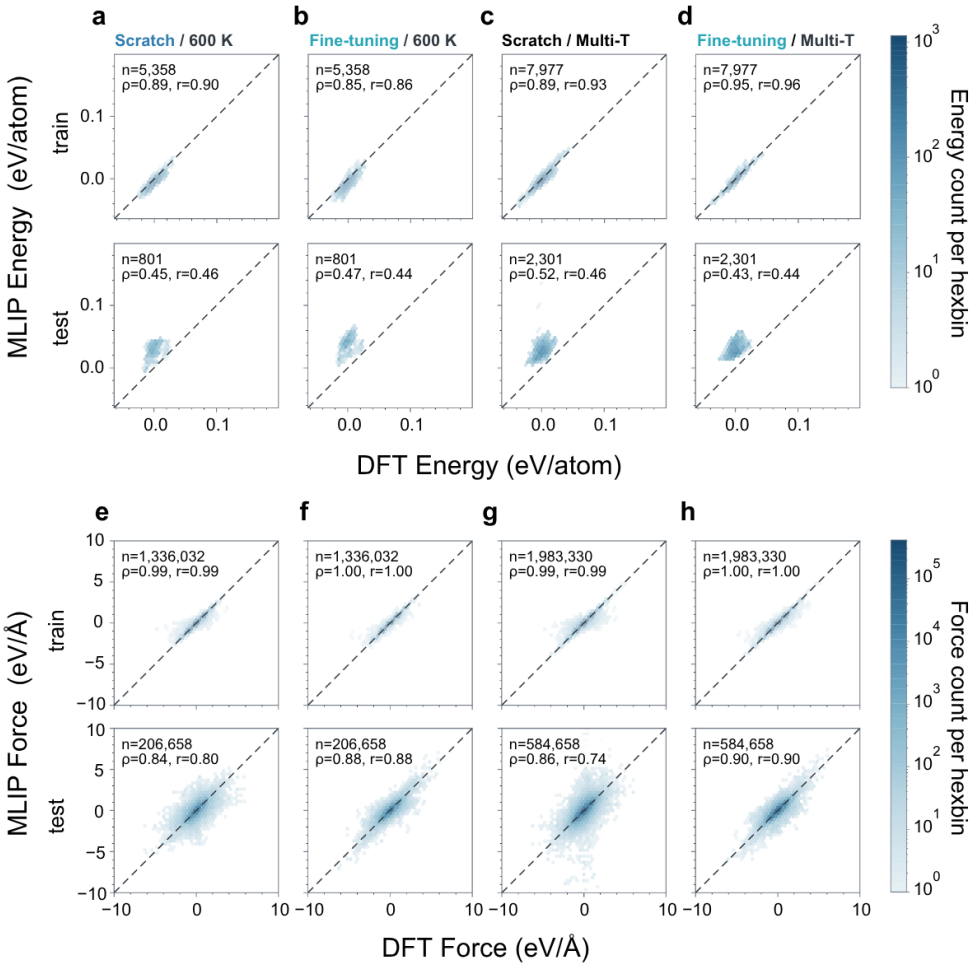


Fig. 2 Training and held-out pointwise agreement across four model conditions. a–d, Energy parity for Scratch-600K, FT-600K, Scratch-MultiT and FT-MultiT, respectively; within each panel, the upper and lower plots show training and test data. **e–h,** The corresponding force-component parity plots in the same model order and row order. Energies are per atom and are centred by subtracting the pooled DFT mean for the corresponding trajectory cluster from both DFT and MLIP values; test-only cluster means are used for visualization only. Forces are uncentred Cartesian components. Predictions from three training seeds are pooled, so the displayed n counts prediction records rather than independent structures: 5,358/801 training/test energy records for each 600 K family and 7,977/2,301 for each MultiT family. Hexbin colour denotes count on logarithmic scales, with separate normalizations for energy and force. Dashed lines indicate identity; ρ and r are Spearman and Pearson correlations. The force display is restricted to ± 10 eV Å^{-1}, leaving 106 Scratch-MultiT test components outside the plotted window; those components remain in all numerical metrics. Uncentred, seed-resolved errors are given in Supplementary Table S1.

2.2 Stable trajectories and accurate dynamics are separate requirements

Trajectory survival differs sharply across the tested routes (Fig. 3a–f). All three zero-shot foundation checkpoints reach the 1 ns target. Among the trained checkpoints, only FT-600K seed 234 reaches that target. The other two FT-600K runs end at 236 and 485 ps, and the FT-MultiT runs end between 237.2 and 286 ps. All six scratch runs reach a scientific stopping threshold between 200.4 and 204.4 ps (Supplementary Table S8). These outcomes use one common starting state and thermostat protocol; they do not estimate failure probabilities over independent thermal replicas. Under this protocol, the zero-shot foundation checkpoints provide longer usable trajectories than most adapted or specialist checkpoints.

Survival alone does not establish dynamical accuracy. The foundation trajectories show greater mobility in the in-plane MSD and a different local Cr environment, even where the all-atom radial distribution functions retain similar principal peaks (Fig. 3c–e). A global structural average can therefore conceal a local dopant discrepancy. Kinetic temperature remains regulated while pressure and other indicators differ across models (Fig. 3a,b).

The truncated trajectories make a common long-time diffusion ranking inappropriate. A short nearly immobile trace is not evidence of a physically small diffusion coefficient when it terminates at a stopping threshold. We use the recorded MD observables to distinguish observed stability, local structure and mobility, and do not infer equilibrium transport coefficients or thermal conductivity from this comparison.

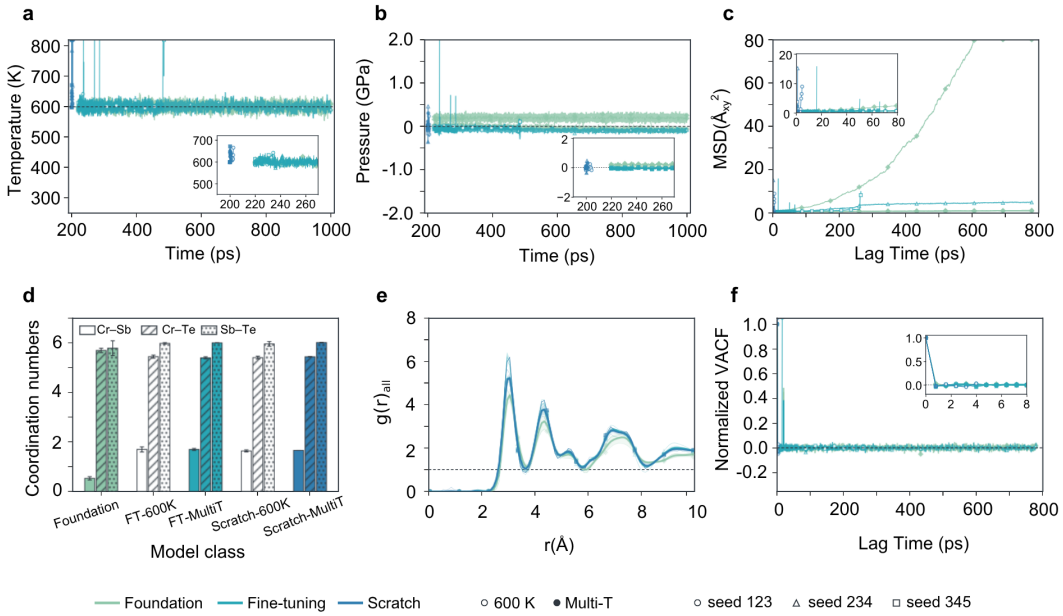


Fig. 3 Molecular dynamics tests the configurations generated by each model. The comparison contains MPA-0, MP-0b3 and OMAT-0 foundation checkpoints and seeds 123, 234 and 345 for FT-600K, FT-MultiT, Scratch-600K and Scratch-MultiT. **a,** Kinetic temperature. **b,** Hydrostatic pressure, $P = -\mathrm{tr}(\sigma)/3$, from recorded stress tensors. **c,** In-plane, multiple-origin MSD with triclinic periodic unwrapping. **d,** Pair-resolved coordination numbers, using fixed Cr–Sb, Cr–Te and Sb–Te cutoffs of 3.2, 4.1 and 4.2 Å, respectively; bars and whiskers summarize the mean and sample standard deviation within each displayed condition. **e,** All-atom RDFs from ordered pair counts and triclinic minimum-image distances; thin traces retain individual models and the summaries use display-only smoothing. **f,** Normalized multiple-origin velocity autocorrelation. Insets expand early-time behaviour. Colours distinguish strategies, open/filled symbols distinguish 600 K/MultiT training, and markers identify training seeds; foundation differences are checkpoint variation. Runs start from a common state at 200 ps and target a total simulation time of 1 ns, using a 1 fs timestep and 200-step output interval. Curves end at their observed horizons; structural summaries use the available, unequal trajectory windows. Only three foundation checkpoints and FT-600K seed 234 reach the target (Supplementary Table S8).

2.3 A correct energy assignment does not guarantee a correct pathway

The migration results depend on both the path class and whether the geometries are fixed or model-relaxed (Fig. 4a–f).

On fixed images, foundation and fine-tuned profiles retain the qualitative shape of the in-gap reference more closely than the legacy scratch profile (Fig. 4a). That relative success does not extend to the layer interior: foundation and fine-tuned profiles assign much larger energies to strongly distorted within-QL images (Fig. 4c; Fig. 5b,c). The seed-resolved matrix confirms that this discrepancy is not confined to a single displayed profile.

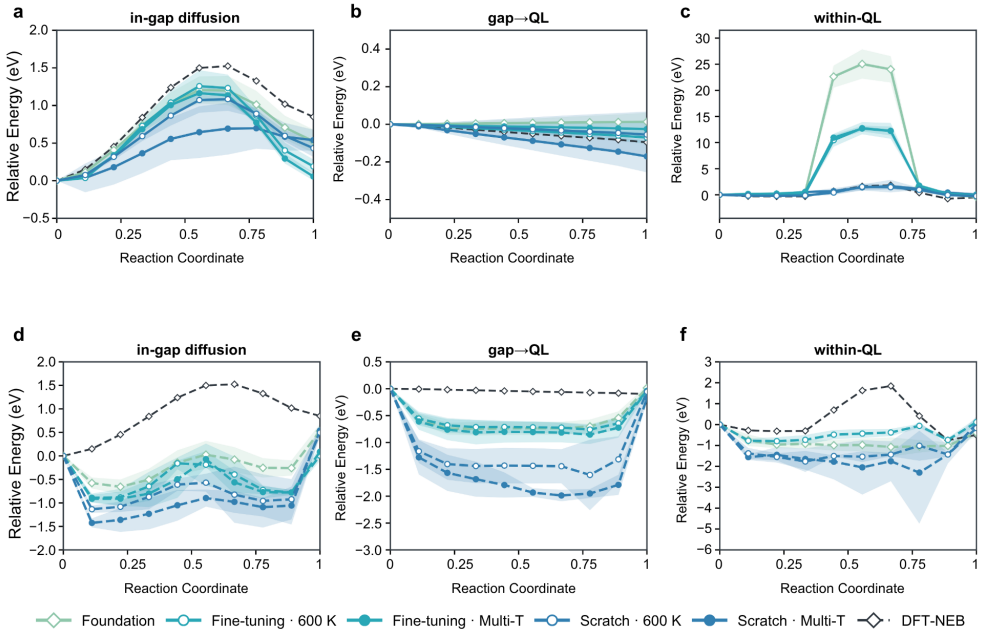


Fig. 4 Fixed geometries and model-relaxed paths reveal different discrepancies. a–c, Energy profiles evaluated on fixed reference images for in-gap, gap-to-QL and within-QL motion, respectively. **d–f,** Corresponding endpoint-initialized, MLIP-relaxed pathway diagnostics. Energies are relative to each profile's first image; the horizontal coordinate is normalized image index and does not imply that optimized paths follow the same geometric arc. Dashed dark curves are the DFT reference profiles. Coloured curves and bands summarize the selected model groups and their pointwise variation. The scratch curve is explicitly the legacy scratch condition, rather than the complete contemporary Scratch-600K/MultiT seed matrix in Fig. 5b–e; ensemble membership is documented in Supplementary Table S3. Missing model profiles indicate unavailable or excluded accepted records. The reference peaks are 1.5243, 0 and 1.8449 eV for the three columns. Their source records retain SCF warnings, so they are image-set energy diagnostics. Self-relaxed profiles are endpoint-only MLIP NEB proxies and do not establish DFT-validated minimum-energy paths.

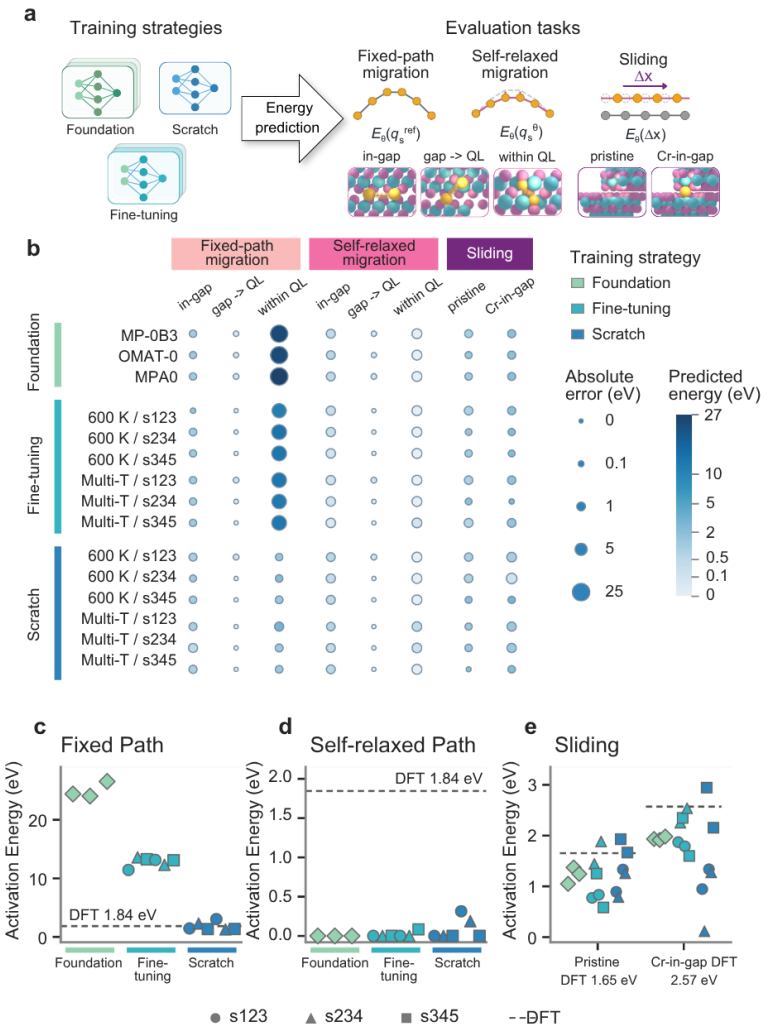


Fig. 5 Reliability depends on the task and evaluation operator. a, Evaluation design: three foundation checkpoints and three training seeds for each of the four trained conditions (scratch or fine-tuned, each using 600 K or MultiT data), evaluated over three fixed migration paths, three self-relaxed migration paths and two fixed sliding paths. **b,** The 120-record source matrix. Colour encodes the predicted forward peak relative to the first image; circle area follows a square-root mapping of the absolute discrepancy from the reference. Colour uses a power normalization with exponent 0.40. A minimum dot remains at zero discrepancy. **c,d,** Within-QL forward peaks for fixed and self-relaxed paths; the reference line is 1.8449 eV and the vertical scales differ. **e,** Fixed-path sliding peaks for pristine and Cr-in-gap systems, with reference peaks of 1.6538 and 2.5708 eV. Shapes identify training seeds or foundation checkpoints. All finite values are retained as diagnostics. Crosses identify five nonconverged records; the remaining source tags comprise 75 finite-diagnostic and 40 optimizer-converged records. A hash-matched raw-record correction changes the Scratch-600K seed-123 within-QL self peak from 0 to 2.351 eV and corrects two convergence labels (Supplementary Sec. 2.2). Neither a small dot nor a small self-relaxed peak certifies a physical barrier. Migration references retain SCF warnings, and fixed sliding endpoints and profiles require separate consistency checks. Numerical values, statuses and the correction audit accompany the figure.

On the selected within-QL image set, the reference peak relative to the first image is 1.8449 eV. In the fixed-path comparison (Fig. 5c), the three zero-shot checkpoints assign 24.04–26.54 eV and the six fine-tuned checkpoints assign 11.44–13.58 eV. Scratch-600K gives substantially smaller absolute reference discrepancies, 0.45 ± 0.05 eV, across its three seeds (Supplementary Table S4). This lower fixed-image discrepancy is reproducible across the tested Scratch-600K initializations; it should not be dismissed as a chance result, although it does not establish a reliable self-generated pathway.[R1-A3] Those same scratch training conditions do not sustain the MD continuation.

The in-gap comparison illustrates why a statement about fine-tuning must identify its comparator (Fig. 5b). FT-600K has a mean absolute reference discrepancy of 0.27 ± 0.21 eV in the seed-resolved comparison, compared with 0.44 ± 0.16 eV for Scratch-600K; the three zero-shot values span 0.31–0.38 eV. The small number of seeds and checkpoint differences do not justify a general ordering from these overlapping ranges. The material, image set and checkpoint lineage must therefore remain attached to each result.

Several model-relaxed profiles have deep intermediate minima or small forward peaks while missing the shape of the reference (Fig. 4d–f). Most within-QL forward peaks collapse towards zero despite the large fixed-image discrepancies (Fig. 5d). Such a reduction is not evidence that relaxation has corrected the energy landscape; it can also result from changing the sampled path, basin or endpoint relationship. A finite or optimizer-converged band still requires structural correspondence and DFT validation before it can support a kinetic interpretation.

The reference quality sets a further limit. The canonical ten-image migration references used in Fig. 4a–c and Fig. 5b–d retain electronic-convergence warnings in their source records. Their reported peaks are therefore diagnostic image-set references, not validated activation energies. A zero forward peak for the downhill gap-to-QL reference likewise tests neither the reverse event nor the complete profile. We retain these probes to expose large disagreements and operator dependence, rather than to infer migration rates.

2.4 Collective sliding tests what local adaptation leaves unconstrained

Relative layer displacement changes registry throughout the structure. It therefore probes a different response from moving a single Cr atom, even though each task occurs in the same host. Figure 5e and Supplementary Fig. S10a–d compare pristine and Cr-containing sliding profiles. The predicted peaks vary substantially across seeds, and some approach a reference peak while failing to reproduce the profile or the nominal endpoint return.

This distinction is particularly visible in the fixed-path curves. The DFT sliding profiles return close to their initial reference energy, whereas several MLIP curves retain an endpoint offset. A matching peak can coexist with an incorrect registry response. Such an offset must first be checked against the atom mapping, periodic displacement and calculator consistency; it is not, by itself, evidence that a model has

lost long-range physics. Similarly, a coordination descriptor or a finite cutoff does not identify which interaction caused the discrepancy.

The joint result is a task-specific trade-off. A training route with a smaller discrepancy on one distorted image set can fail under its own dynamics. A checkpoint with a long-lived trajectory can misassign a migration energy. Sliding adds a profile and endpoint-consistency requirement that neither of those tests supplies. These observables should remain separate until a particular application provides a reason to combine them.

2.5 Projection geometry is a diagnostic, not a mechanism

The 15 model conditions occupy visibly different regions in the displayed trajectory projections (Fig. 6a–c). The projected features describe recorded energies, forces and structural summaries; they are not native MACE hidden states, so their separation cannot establish distinct learned physical mechanisms.[R1-A6] The sampled populations also differ in duration because the scratch trajectories terminate early.

There is a quantitative reason to resist a stronger reading. The unprojected output-feature silhouette is 0.063 and becomes −0.059 after feature standardization, compared with 0.117 in the displayed two-dimensional t-SNE projection (Fig. 6d; Supplementary Fig. S12a).[R1-C11] Repeating t-SNE across three perplexities and three nominal seeds gives projected silhouettes of 0.079–0.133 (Supplementary Fig. S12b). Raw model energies have different reference offsets, and the SOAP summaries were fitted with separate PCA bases. Separation on the common 30-bin neighbour histogram is 0.056 before standardization. None of these values supports a claim of disjoint local-geometry manifolds.

The useful role of the projections is narrower. They identify trajectory populations and motivate checks of the underlying structures, scales and failure horizons. Their connection to a physically relevant error must be established with the process observables themselves, rather than inferred from the distance between coloured clusters.

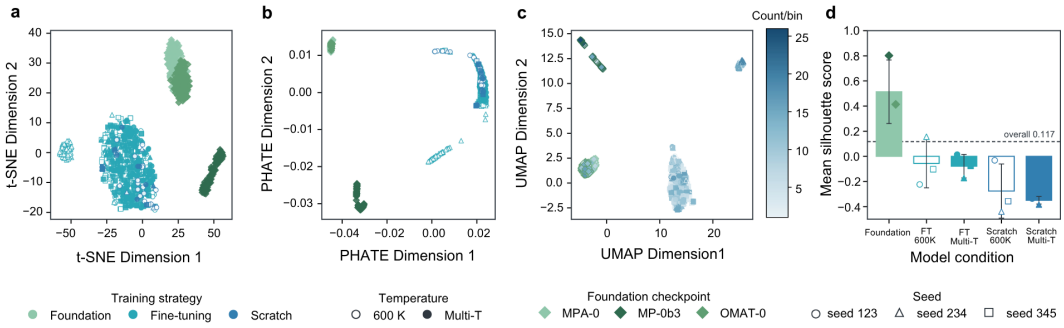


Fig. 6 Trajectory-feature projections across the 15-model MD comparison. a, t-SNE with perplexity 30 and random state 42. **b,** PHATE with random state 42. **c,** UMAP with 15 neighbours, minimum distance 0.1 and random state 42; colour intensity indicates local two-dimensional bin count on a 45-bin histogram, not energy or physical probability. **d,** Per-condition means of model-specific silhouette scores computed in the t-SNE plane, with individual checkpoint/seed symbols and a dashed overall value of 0.117. These are projection-space scores, not original-space separability or independent thermal-replica uncertainty. The input contains 927 frames, with 4–101 frames per model, and combines energies, Cartesian forces, neighbour histograms and per-trajectory SOAP/PCA summaries. Model energy offsets, feature scale, separate SOAP PCA bases and unequal failure horizons limit geometric interpretation. Original-space and projection-sensitivity checks on the same dataset are reported in Supplementary Fig. S12a,b and Supplementary Table S9. Constant columns are excluded from the quantitative distance analysis.

2.6 Error surrogates reveal associations and their instability

Feature rankings vary across training seeds and model conditions (Fig. 7a,b). Te–Te and Cr-containing descriptors appear among the leading fine-tuned features, while scratch conditions emphasize a different combination of host and dopant environments. Several between-seed standard deviations are comparable to or larger than the corresponding means. This variability argues against identifying a single dominant Cr–Cr channel or assigning the observed pathway behaviour to one internal feature.

The current cross-validation diagnostic gives R^2 values of 0.981 ± 0.003 for FT-600K, 0.973 ± 0.003 for FT-MultiT, 0.976 ± 0.006 for Scratch-600K and 0.952 ± 0.022 for Scratch-MultiT (Supplementary Table S11). These values describe the five-fold diagnostic surrogate across the three training seeds, not independent extrapolation to a new trajectory. The final, higher-capacity TreeSHAP estimator is fitted on the full common sample; its in-sample fit does not validate generalization.

A supplementary coordinate-perturbation study tests the local relaxation response of native MACE inputs across six dopant–host combinations. It supplies a physical-input robustness check, but does not isolate a SOAP channel or establish causal attribution for migration errors.

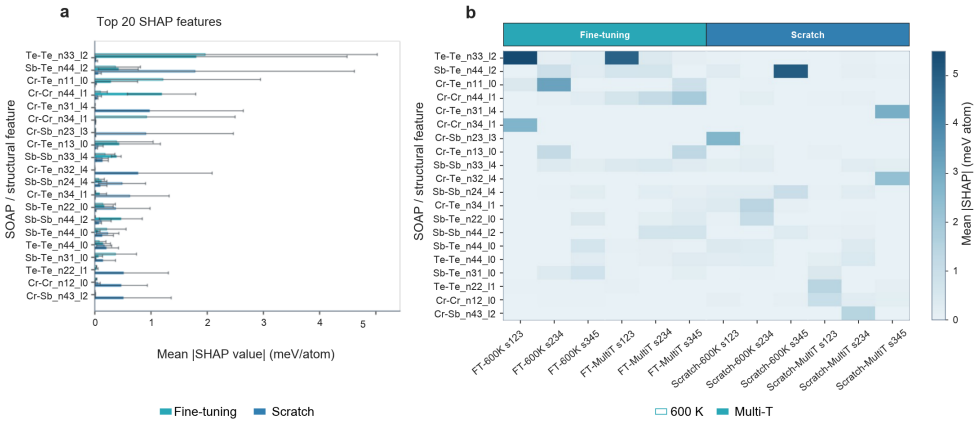


Fig. 7 Structural predictors of energy error depend on model condition and training seed. TreeSHAP explains gradient-boosted surrogates of the absolute energy error per atom on one common 267-frame test trajectory. **a,** Mean absolute SHAP values for the 20 features selected by the shared global ranking, with mean and sample standard deviation over training seeds 123, 234 and 345 within each of four conditions. Strategy colour distinguishes fine-tuning and scratch training; open/filled bars distinguish 600 K and MultiT. **b,** The same 20 feature rows aligned across all 12 model–seed columns on a shared meV atom^{-1} scale. Species pairs and n_1n_2, l identify SOAP power-spectrum channels; they are descriptor indices, not isolated bonds or MACE neurons. Surrogates use 407 computed features and frame-aligned errors. Fidelity of the separate five-fold diagnostic estimator is given in Supplementary Table S11; it does not make the full-data TreeSHAP fit an out-of-sample causal explanation.

3 Discussion

When does a foundation model help? In this study, the answer changes with the observable. The tested zero-shot checkpoints sustain the longest MD continuations, while target adaptation reduces held-out force error and improves selected fixed-image energy comparisons. Neither benefit supplies the other. Within-QL distortions expose large energy discrepancies in foundation and adapted models, while scratch models with smaller discrepancies fail rapidly in MD. The central lesson is that a transferable prior and a trustworthy simulation are different objects: the former becomes useful only where the latter has been established for the intended process.

These results suggest a practical order for making that decision. A candidate model should first be checked against the target electronic-structure reference on common configurations, including the coordination changes that matter to the process. It should then be allowed to generate short paths or trajectories, with convergence, geometry and stopping criteria inspected directly. Collective displacements and endpoint-return checks can test responses that local thermal sampling leaves unconstrained. Passing one stage provides evidence for that stage; it does not certify the remaining observables.

Distribution shift is a useful way to organize this reasoning, but its meaning must be specific. The adaptation dataset can miss a process-relevant geometry even within the same chemistry and temperature range. Model-driven trajectories can create a further shift by visiting configurations absent from the reference set. Differences between the

pretraining and target electronic-structure protocols can add calibration errors. We do not have a complete, matched audit of the foundation pretraining distributions, so the present results do not identify a numerical pretraining-distance threshold or prove that distribution shift is the sole cause of failure.

This distinction also gives active learning a more precise objective. A promising next step is to acquire DFT labels where process-relevant displacements or short model rollouts reveal large errors, unstable forces or unresolved reference profiles. Updating a model should then be judged by improvement on held-out process tests while rechecking the behaviours that previously worked. Such a loop could guide the choice between adaptation and a specialist, and constrain how adaptation is performed. Feature attributions or model disagreement may help nominate configurations, but their acquisition value requires a prospective comparison under a matched DFT budget.

The conclusions concern the tested MACE routes in Cr-doped Sb_2Te_3; additional dopants, hosts and controlled architecture-matched comparisons are needed before treating the workflow as a general model-selection standard.[R1-A4] The diagnostic references with convergence warnings further preclude quantitative kinetic claims. Within those limits, the tests answer a useful experimental question: a foundation model helps when it supplies accuracy or robustness that survives the specific process being simulated, and fine-tuning helps only when that benefit survives the same scrutiny.

4 Methods

4.1 Reference data and model lineages

The reference calculations and AIMD data use PBE-based Quantum ESPRESSO for Cr-doped Sb_2Te_3 [68–70]. The 600 K target pool matches the thermostat temperature used for the MD comparison. The MultiT pool broadens thermal coverage and tests whether training across temperatures carries over to the target-temperature trajectories and process configurations. Because the two pools differ in size and test composition, the 600 K–MultiT contrast is not a temperature-only experiment and cannot isolate the causal effect of broader sampling. Path-specific computational settings and convergence states are distinguished in Supplementary Sec. 1; warning-bearing migration references are not combined with converged paths to obtain a pooled kinetic score.

The contemporary pointwise and MD comparison contains the MPA-0, MP-0b3 and OMAT-0 zero-shot foundation checkpoints and three seeds for each of four trained conditions: Scratch-600K, FT-600K, Scratch-MultiT and FT-MultiT. Fine-tuned grouped-A2 models use an OMAT-FT starting checkpoint; scratch models are the registered $L_{\max} = 1$ implementations. Legacy pathway checkpoints are identified separately in Fig. 4a–f and Fig. 5b–e. Figure 5a and Supplementary Fig. S1a,b summarize the evaluation and training-route designs.

Training, validation and test partitions are assigned by complete source trajectory for the grouped comparisons, so adjacent frames from one source do not cross partitions.[R1-C10] Within each temperature regime, scratch and fine-tuned models use the same target-data split and are evaluated with seeds 123, 234 and 345. The 600 K

pair uses 1,786/311/267 training/validation/test configurations; the MultiT pair uses 2,659 training and 767 test configurations. This design supports a direct route comparison at fixed 600 K target data and a separate comparison under broader thermal coverage. An additional overlap audit of 25 legacy migration images finds no exact or threshold-defined SOAP near-duplicate in the declared training files; its scope is that audited image set, not every subsequently assembled path.[R1-A2] The separate 89-configuration Scratch-5% control is a data-volume sensitivity check, rather than a data-matched estimate of the effect of pretraining.[R1-A5] Architecture and optimization remain part of each route, so the route comparison does not isolate pretraining alone.

The fine-tuning provenance audit finds no NEB-derived or unclassified all-zero-force training frame among 9,163 audited records; the nine exact-zero-force records are labelled one-atom energy-reference structures.[R1-C8] Missing-force NEB exports are excluded from force-supervised training. The supporting audit retains the source-file identities and does not imply that every later campaign has the same training provenance.

4.2 Process observables and uncertainty

Fixed-path scoring evaluates every model on the same reference geometries, whereas self-relaxed scoring uses an endpoint-initialized NEB path whose intermediate geometries are moved by that model's forces. These operators test different properties because self-relaxation changes both the energy function and the sampled configurations. For a reference image set $\Gamma_{\mathrm{ref}} = \{X_i\}$, the fixed-path forward peak is

$$B_M^{\mathrm{fixed}}(\Gamma_{\mathrm{ref}}) = \max_i \{E_M(X_i) - E_M(X_0)\}. \tag{1}$$

The diagnostic discrepancy is $|B_M^{\mathrm{fixed}} - B_{\mathrm{ref}}|$. It is an activation-energy error only when the electronic reference, pathway optimization and endpoint definition support that interpretation. A self-relaxed value uses $\Gamma_M = \mathrm{NEB}_M(X_0, X_1)$ instead. A lower self-relaxed peak therefore does not establish a more accurate physical barrier without structural correspondence and reference convergence. The fixed and self-relaxed operators measure energy calibration on a supplied path and the outcome of model-driven path optimization, respectively; their peak energies cannot be interchanged as evidence for the same physical barrier.[R1-A1] Full profiles and status records are retained alongside peak values.

Each MD run starts from the same 2,050-atom state at a total simulation clock of 200 ps and proceeds toward a total simulation time of 1 ns.[R1-C9] The timestep is 1 fs, trajectory output is every 200 steps, the thermostat target is 600 K, the Langevin friction parameter is 0.01 and the MD seed is 1100. Time is reconstructed from recorded steps, and each trajectory is retained through its actual endpoint; stopped runs are not extended or treated as completed 1 ns trajectories. Sample standard deviations across training seeds describe checkpoint variation, while variation among foundation models is across distinct checkpoints. Neither estimates uncertainty over independent starting-state or velocity replicas [71].

4.3 Diagnostic analysis and reproducibility

The trajectory-feature analysis embeds 927 sampled frames from 15 MD model conditions with t-SNE, PHATE and UMAP. Quantitative distances exclude constant columns and contain no force-sensitivity channel; original-space scores are calculated before and after standardization, and t-SNE is repeated at perplexities 5, 30 and 50 with nominal seeds 0, 42 and 123. A common neighbour-histogram analysis avoids treating independently fitted SOAP PCA coordinates as one shared physical basis. The feature analysis describes output features, not internal model activations, and the quantitative distance analysis excludes constant columns.[R1-C12]

The SHAP analysis uses a separate common 267-frame, 86-atom held-out trajectory and all four trained conditions with three seeds each. The target is absolute energy error per atom; inputs are 390 periodic SOAP features and 17 structural descriptors. A gradient-boosted regressor with 200 estimators, learning rate 0.05, depth 8, subsampling 0.8 and random state 42 supplies the full-sample TreeSHAP values. A separate 50-estimator, depth-4 surrogate uses shuffled five-fold validation to report diagnostic fidelity. The frames belong to one trajectory, so these folds do not measure generalization to an independent trajectory. Full definitions and per-seed metrics are provided in Supplementary Sec. 6.

Data availability

The figure-associated numerical tables, model and dataset identities, convergence-status records and current diagnostic outputs accompany the submission source package. The project repository is https://github.com/yicao-elina/MigrationBench. Figure-specific source identities are provided with the Supplementary Data; a family label alone is not a checkpoint identifier.

Code availability

The accompanying source package includes the executed scripts and frame-aligned inputs for the current SHAP analysis, the current feature audit, and the scripts and compact source tables used to assemble the figure evidence.[R1-C13]

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Author contributions

Y.C.: conceptualization, methodology, data curation, DFT and MD simulations, analysis, visualization and writing. P.M.: literature review, conceptualization and writing. P.C.: conceptualization, project administration, supervision, funding acquisition and writing.

Competing interests

The authors declare no competing interests.

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